DAY 20 | POST-READ

Git, Pipelines and Docker: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Git areas	Working folder, staging (git add), repository (git commit), remote (git push).
Branches and PRs	One branch per task; small, explained, green PRs; branch policies protect main.
Pipeline levels	Pipeline, stages, jobs, steps. dependsOn for order; condition for main only.
Security gates (NEW v4)	bandit (SAST), pip-audit and npm audit (dependencies), secret scan, image scan.
Fail, remediate, re-run (NEW v4)	Triage, fix, prove with a test, re-run, log in vulnerability report v2.
AI PR review (NEW v4)	Codex comments on the diff; a human approves the merge.
Docker terms	Dockerfile (recipe), image (package), container (running), registry (storage).
Image hardening (NEW v4)	Slim base, non-root user, no secrets in layers, pinned versions, scanned in CI.
Full stack in containers	Env config, /health and /ready, multi-stage React build, docker compose.

Part B: Going beyond today

1. Good commit messages

A short summary line, a blank line, then why. Your future self will thank you.

```
Add priority filter to ticket list

Agents asked to focus on urgent work. Adds ?priority= to GET /tickets
and a dropdown in the dashboard. Closes #42.
```

2. Layer caching

Docker caches each instruction. Copy requirements.txt and install before copying code, so code edits do not reinstall every package. Builds go from minutes to seconds.

3. Pre-commit hooks

Run ruff and a secret scan on your laptop before every commit. A secret that never reaches the remote never needs rotating.

4. A leaked secret is a burnt secret

Deleting a key from the latest commit is not enough: it stays in history and may already be copied. Revoke and rotate it first, then clean up.

5. Pipeline templates and variables

Large teams reuse steps with templates and keep settings in variable groups, with secrets linked from Azure Key Vault. You meet Key Vault tomorrow.

Part C: Check yourself

- What is the difference between git add and git commit?
- Name the four security gates from today and what each one catches.
- Why must a human, not Codex, approve the merge?
- Name three ways to harden a Docker image.
- What does -p 8000:8000 mean in docker run?

Part D: Practice ideas for tonight

- Add a .dockerignore and compare image sizes before and after.
- Plant a fake key in a practice repo and make your secret scanner fail the build.
- Ask Codex to review one of your own pull requests and list which comments you accepted and why.

Coming up on Day 21: Azure App Service, Container Apps, continuous delivery, Key Vault and managed identity.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today